

Homework (Jalapeno)

- Q1.** The ancient civilization of Babylon used $x \cdot y = 12$ and $x + y = 7$ to build their ziggurats. What is the solution to this system?
(A) $x = 3, y = 4$
(B) $x = 2, y = 5$
(C) $x = 4, y = 3$
(D) $x = 1, y = 6$
(E) $x = 6, y = 1$
- Q2.** In ancient Egypt, the equation $2x + 3y = 12$ was used to measure the height of pyramids. If $x = 3$, then what is y ?
(A) $y = 2$
(B) $y = 1$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$
- Q3.** A ancient Greek potter uses $x + 2y = 8$ to create symmetrical vases. If $y = 2$, then what is x ?
(A) $x = 4$
(B) $x = 3$
(C) $x = 2$
(D) $x = 1$
(E) $x = 0$
- Q4.** The Mayans used $3x - 2y = 5$ to calculate the movement of celestial bodies. What is the y -intercept of this line?
(A) $y = -\frac{5}{2}$
(B) $y = 5$
(C) $y = 0$
- Q5.** The ancient Chinese used $x - y = 2$ to balance their elements. What is the slope of this line?
(A) $m = 1$
(B) $m = -1$
(C) $m = 2$
(D) $m = -2$
(E) $m = 0$
- Q6.** The ancient Romans used $2x + y = 7$ to build their aqueducts. What is the slope of this line?
(A) $m = 2$
(B) $m = -2$
(C) $m = -\frac{1}{2}$
(D) $m = \frac{1}{2}$
(E) $m = 0$
- Q7.** The Aztecs used $x + y = 5$ to create their calendars. What is the x -intercept of this line?
(A) $x = 5$
(B) $x = -5$
(C) $x = 0$
(D) $x = 1$
(E) $x = -1$
- Q8.** The ancient Sumerians used $y = 2x - 3$ to measure the growth of their crops. What is the y -intercept of this line?
(A) $y = -3$
(B) $y = 3$
(C) $y = 2$
(D) $y = -2$
(E) $y = 0$

Open Ended Questions (FRQ)

- Q9.** Graph the system $x + y = 4$ and $x - y = 2$. Identify the intersection point and explain its significance in the context of ancient trade routes.
- Q10.** The ancient Egyptians used the system $2x + y = 7$ and $x - 2y = -3$ to design their temples. Solve this system by graphing and explain the historical context of this problem.

Chef's Tips

- Tip 1: Ensure students understand the historical context of each problem.
- Tip 2: Emphasize the importance of graphical solutions in real-world applications.
- Tip 3: Encourage students to use different methods to solve systems, such as substitution and elimination.